

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 2
of 40 — every set uses different data)

Decode the Chart

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For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

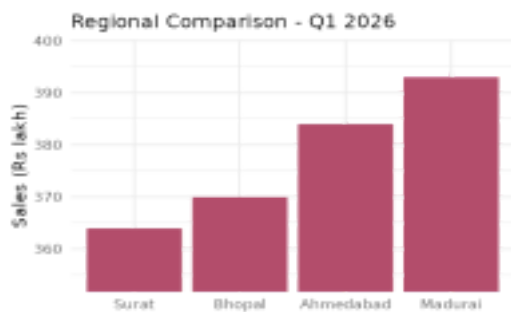


Chart 1

Budget Allocation by Department (11 categories)



Chart 2

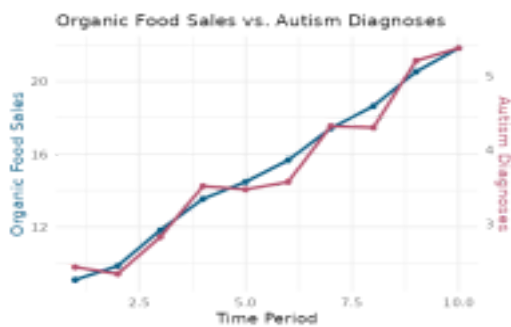


Chart 3

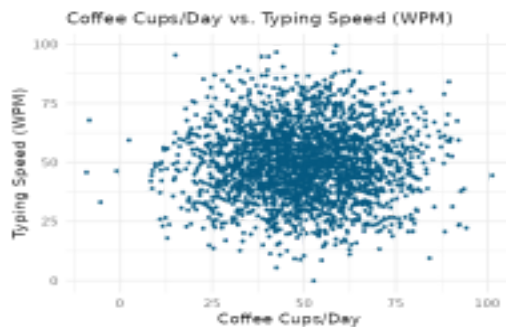


Chart 4

Chart 1 — Regional Comparison

What is it trying to tell you?

The chart compares the values of different regions. The values increase in regular intervals of about **10 units**, so the graph is mainly showing the regional differences and indicating which region has the highest and lowest value.

Do you trust it?

Yes, I mostly trust the information shown by the graph.

Why?

The values are given clearly and the difference between the regions follows a consistent **10-unit interval**, so the actual numerical comparison is understandable. However, the Y-axis does not start from zero, which makes the visual difference between the bars appear larger than the actual difference. Therefore, I would trust the **given values**, but I would be careful about judging the size of the difference only from the height of the bars.

Chart 2 — Budget Allocation

What is it trying to tell you?

The chart shows how the total budget is divided among different categories. It helps us understand which areas receive a larger portion of the budget and which receive a smaller portion. The larger slices represent categories with higher budget allocations, while the smaller slices represent lower allocations.

Do you trust it?

I can trust the overall budget distribution, but I would be cautious when comparing individual categories.

Why?

The budget values are represented as portions of the total, so the overall idea of how the budget is allocated can be understood. However, because the pie chart contains many categories, several slices are very similar in size and are difficult to compare accurately. It is therefore possible to understand the general distribution, but identifying the exact ranking of smaller categories is difficult. A bar chart would make the comparison more precise.

Chart 3 — Organic Food Sales

What is it trying to tell you?

The chart compares **organic food sales** with another quantity over the same period. The two lines appear to increase and decrease in a similar pattern, suggesting that the two quantities may be related.

Do you trust it?

I would trust the individual sales values, but I would not completely trust the apparent relationship between the two quantities.

Why?

The values shown for organic food sales can still be read from the graph. However, the two variables are displayed using **different Y-axis scales**. Because each axis has its own scale, the lines can be made to appear as though they move together even when the actual changes may be very different. Therefore, the graph is useful for observing the individual trends, but the visual similarity should not by itself be taken as proof of a relationship.

Chart 4 — Scatter Plot

What is it trying to tell you?

The scatter plot is trying to show the relationship between the two variables by plotting individual observations. The concentration of points helps us identify whether the variables appear to have a particular pattern or relationship.

Do you trust it?

I can trust that the data points represent the observations, but I would be cautious about judging the exact distribution from the graph.

Why?

There are a very large number of points, and many points overlap one another. Because of this overplotting, dense areas appear as solid regions and individual observations become difficult to distinguish. This makes it harder to identify where the data is actually concentrated and whether there are smaller patterns or outliers. Using transparency or a 2D histogram/hexbin plot would make the distribution clearer.

Chart	Graph Type	Example	Advantage	Disadvantage	Trust?	Reason
1	Bar Chart	Regional Comparison	Easy to compare values between regions and identify highest/lowest values.	Truncated Y-axis can make the differences look larger than they actually are.	Yes, with caution	The values have a consistent 1 difference, so the numerical comparison is meaningful. However, the visual size of the bars should be judged alone because the Y-axis does not start at zero.
2	Pie Chart	Budget Allocation	Clearly shows how the total budget is divided among different categories.	Too many slices make similar categories difficult to compare.	Partially	The overall allocation can be understood, but comparing individual slices accurately is difficult because there are many categories.

3	Dual Y-axis Line Chart	Organic Food Sales	Can show the movement of two quantities over the same period.	Different Y-axis scales can make two unrelated trends appear correlated.	Partially	The individual values/trends can be observed, but the apparent relationship between the two variables should not be assumed to be real just because the lines move similarly.
4	Scatter Plot	Relationship between two variables	Shows individual observations and can reveal patterns, trends, clusters and outliers.	Too many overlapping points make the distribution difficult to see.	Partially	The points represent the data, but heavy overlap makes it difficult to accurately interpret the concentration and pattern of the observations.